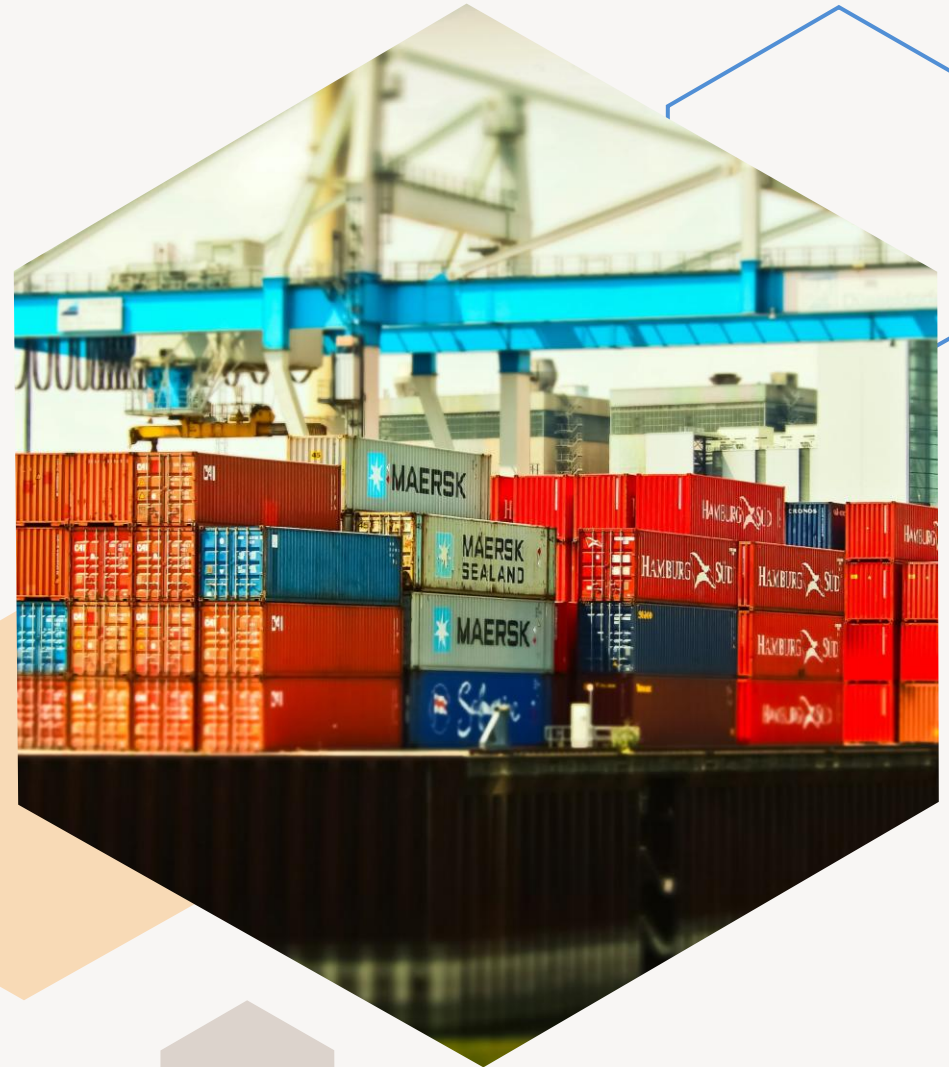


Supply Chain Performance Dashboard & Strategic Analysis 2019

E-Commerce Operations Analytics & Forecasting

Presenter:

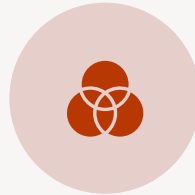
- Chukwudi Ogbuta



External Analysis - Porter's Five Forces



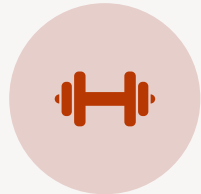
Competitive Rivalry (HIGH): Major players (Amazon, Alibaba, eBay) compete on price, delivery speed, product range



Supplier Power (MODERATE): Multiple suppliers available, but logistics providers have leverage



Buyer Power (HIGH): Low switching costs, price-sensitive customers, easy comparison shopping

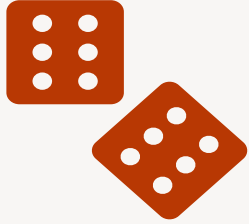


Threat of New Entrants (MODERATE): Low barriers to entry but high cost to scale and compete



Threat of Substitutes (LOW): Traditional retail declining, e-commerce is dominant channel

Data Model & Methodology



Data Model Structure:

Fact Table: FactSales (180,519 transactions with measures: Sales, Profit, Quantity, Discounts)

7 Dimension Tables: DimDate, DimCustomer, DimProduct, DimDepartment, DimShipping, DimCustomerLocation, DimOrderLocation

Relationships: Many-to-One via DateKey, Customer ID, Product ID, Department ID, Location Keys, Shipping Key



Design Rationale:

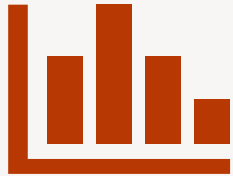
Star Schema chosen over Snowflake for faster query performance and simpler relationships

Separated customer location vs order location to track billing vs shipping patterns

Date dimension enables time-based analysis (YTD, QoQ growth, forecasting)

Clean data: Removed 86%+ null columns, standardized geographies, converted date formats

Dashboard Design & Analytical Framework



Dashboard Architecture:

Page 1 - Descriptive (What happened?): KPI cards, market performance, category sales, delivery status distribution

Page 2 - Diagnostic (Why did it happen?): Late delivery analysis, profit margins, sales-profit correlation, segment performance

Page 3 - Predictive (What will happen?): Trend analysis with 2019 forecasts, QoQ growth tracking, YTD metrics

Page 4 - Prescriptive (What should we do?): 2019 targets, market-specific goals, strategic recommendations

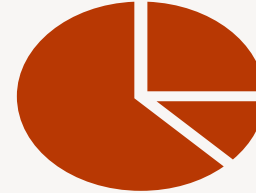


Chart Selection Rationale:

Bar charts for comparisons (markets, categories) - easy to rank

Line charts with forecasting for trend analysis - shows trajectory

Donut charts for proportional breakdowns - parts of whole

Tables for detailed metrics - precise numbers for decision-making

Gauge for goal tracking - visual target vs actual

Internal Analysis- KPIs & 2019 Objectives



2015–2017

Key Performance Indicators

- Total Sales: \$33.05M (peaked 2015: \$11.1M, declined to \$10.6M by 2017)
- Total Profit: \$3.97M (~\$1.3M annually sustained despite sales decline)
- Late Deliveries: 54.82% (critical operational weakness)
- Profit Margin: ~11% (stable but needs improvement)



2019

Strategic Objectives

- Sales Target: \$15M (41% growth)
- Profit Target: \$1.8M (38% increase)
- Delivery Target: <40% late deliveries (15% improvement)
- Target Margin: 12% (margin expansion through pricing optimization)

Sales & Operations Overview

Year
All

Market
All

Category Name
All

Total Revenue (Net Sales)

\$33.05M

Number of Customers

20.65K

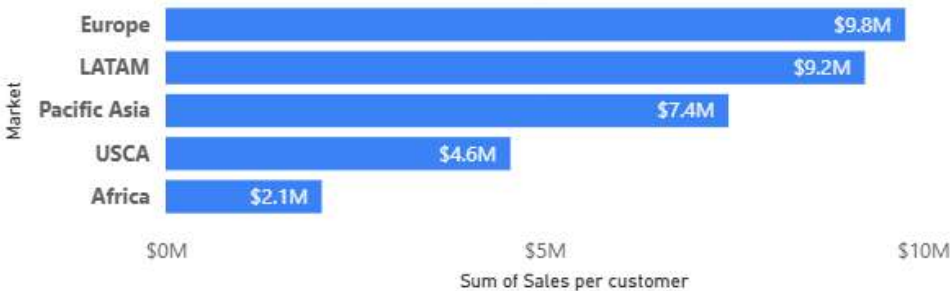
Total Orders

65.75K

Total Profit

\$3.97M

Sales Performance by Market



Top Product Categories



Delivery Status Distribution



Sales by Customer Segment



Key Insight:

Europe leads sales (\$9.8M) | 54.8% late deliveries need improvement | Consumer segment = 52% of revenue

Performance Analysis

Year

All

Market

All

Category Name

All

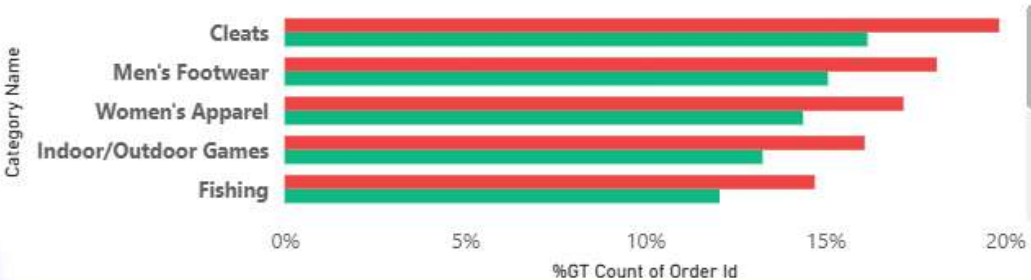
Late Delivery % by Market

Delivery Performance ● Late ● On Time



Top 10 Late Delivery % by Category

Delivery Performance ● Late ● On Time



Sales vs Profit Correlation



Top 10 Profit Margin (%) by Category



Top 10 Products by Profit



Customer Segment Performance

Customer Segment	Sum of Sales per customer	Sum of Order Profit Per Order	Average of Sales pe
Consumer	\$17,164,278.37	\$2,073,487.67	
Corporate	\$10,030,649.90	\$1,202,574.96	
Home Office	\$5,859,474.11	\$690,840.34	
Total	\$33,054,402.38	\$3,966,902.97	

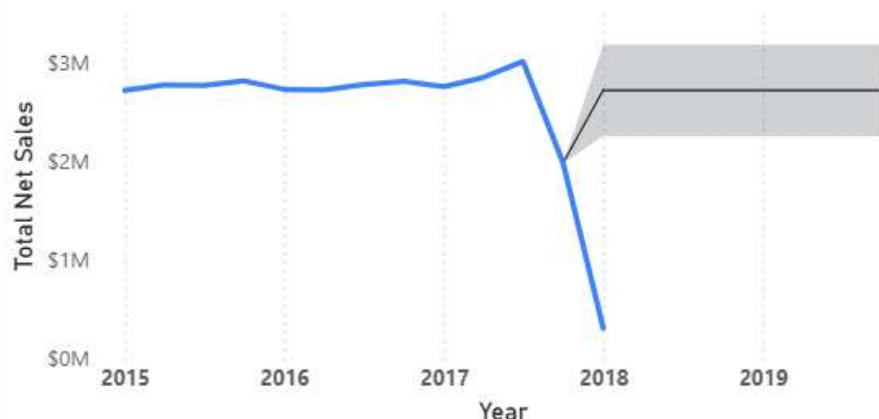
Diagnostic Insight: Europe leads late deliveries (15.5%) | High sales = lower margins | Golf Bags most profitable (19%) | Consumer segment dominates revenue

Trends & Forecasting

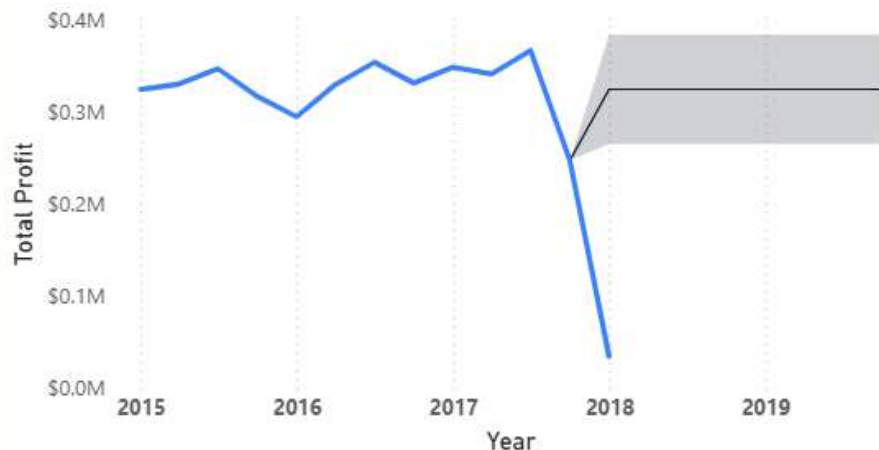
Year

All

Sales Trend Over Time + 2019 Forecast



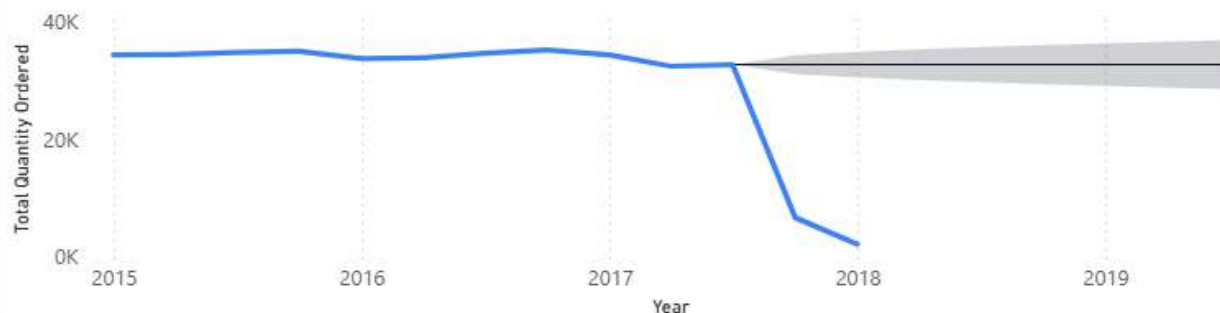
Profit Trend Over Time + 2019 Forecast



Quarterly Performance Analysis

Year	Quarter	Sum of Sales per customer	Previous Quarter Sales	QoQ Growth %	YTD Sales	YTD Profit
2015	Q1	\$2,722,444.70			2,722,444.70	324,579.61
2015	Q2	\$2,776,186.23	2,722,444.70	1.97	5,498,630.93	654,958.35
2015	Q3	\$2,772,652.03	2,776,186.23	-0.13	8,271,282.96	1,002,030.12
2015	Q4	\$2,818,260.59	2,772,652.03	1.64	11,089,543.55	1,318,856.90
2016	Q1	\$2,731,888.11	2,818,260.59	-3.06	2,731,888.11	294,850.43
2016	Q2	\$2,726,919.15	2,731,888.11	-0.18	5,458,807.26	624,739.24
2016	Q3	\$2,781,996.97	2,726,919.15	2.02	8,240,804.23	978,795.21
2016	Q4	\$2,815,192.53	2,781,996.97	1.19	11,055,996.76	1,310,119.07
2017	Q1	\$2,759,687.34	2,815,192.53	-1.97	2,759,687.34	348,643.82
2017	Q2	\$2,852,963.22	2,759,687.34	3.38	5,612,650.56	690,018.81
2017	Q3	\$3,016,043.04	2,852,963.22	5.72	8,628,693.60	1,057,009.06
2017	Q4	\$1,982,216.41	3,016,043.04	-34.28	10,610,910.01	1,304,085.11
2018	Q1	\$297,952.05	1,982,216.41	-84.97	297,952.05	33,841.89
Total		\$33,054,402.38			297,952.05	33,841.89

Order Volume Trend + 2019 Forecast



Predictive Insight: Sales peaked 2015 (\$11.1M), declined to \$10.6M by 2017 | Profit sustained at ~\$1.3M annually despite sales decline - margin improvement | 2018 incomplete (Q1 only: \$298K) | 2019 forecast: ~\$2.7M/quarter suggests flat continuation | Q4 2017 dropped 34% QoQ - investigate cause

Strategic Goals - 2019 Targets

15.00M

2019 Sales Target

1.80M

2019 Profit Target

12.00

Target Profit Margin %

Delivery Performance Goal



Market Growth Strategy & Targets

Market	2017 Sales	2019 Market Target	Gap to Close	Projected Growth %
Africa	223,838.39	316,426.76	92,588.37	41.36
Europe	4,670,280.72	6,602,092.62	1,931,811.91	41.36
LATAM	4,677,604.50	6,612,445.82	1,934,841.32	41.36
Pacific Asia	1,018,672.00	1,440,034.83	421,362.83	41.36
USCA	20,514.40	28,999.96	8,485.56	41.36
Total	10,610,910.01	15,000,000.00	4,389,089.99	41.36

Strategic Recommendations:

Revenue Growth: Ambitious 41% growth target requires aggressive market expansion | **Priority Markets:** Europe & LATAM need \$1.9M increase each - largest opportunity | **Operational Excellence:** Reduce late deliveries from 55% to 40% through logistics optimization | **Margin Improvement:** Achieve 12% profit margin (up from 11%) through pricing strategy and cost control | **Quick Wins:** Focus on high-margin categories (Golf Bags 19%, Fitness 17%) for profit acceleration